

Disproof of Conjecture 00000001132

The structure group of $T(G/B)$ always reduces to a maximal torus:
the rank-1 counterexample $G = \mathrm{SL}_2$

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Abstract

Conjecture 00000001132 states: “*The structure group of the tangent bundle of the complete flag variety reduces to a maximal torus if and only if G is a torus (a characterization of tangent-bundle triviality).*” We disprove it with $G = \mathrm{SL}_2$: the flag variety is $G/B \cong \mathbb{P}^1$, the tangent bundle $T\mathbb{P}^1 \cong \mathcal{O}(2)$ is a complex line bundle whose structure group is $\mathbb{C}^* = \mathbb{G}_m$ — *already* a torus — while SL_2 is not a torus. The mechanism is checked end-to-end by a zero-axiom Lean 4 certificate in the discrete model $\mathrm{SL}_2(\mathbb{F}_2)$.

1 The conjecture

The structure group of the tangent bundle of the complete flag variety reduces to a maximal torus if and only if G is a torus (a characterization of tangent-bundle triviality).

We exhibit G with a reduction of the structure group of $T(G/B)$ to a maximal torus, but with G not a torus, so the “ $\Leftarrow$ ” (indeed the “ $\Rightarrow$ ”) direction fails.

2 The counterexample $G = \mathrm{SL}_2$

Let $G = \mathrm{SL}_2(\mathbb{C})$, B the upper-triangular Borel, $U \leq B$ its unipotent radical, $T = B/U \cong \mathbb{C}^*$ the maximal torus $\{\mathrm{diag}(t, t^{-1})\}$.

1. **Flag variety and tangent bundle.** $G/B \cong \mathbb{P}^1(\mathbb{C})$ and $T_{\mathbb{P}^1} \cong \mathcal{O}(2)$, a complex *line* bundle. As a $\mathrm{GL}_1(\mathbb{C})$ -bundle its structure group is $\mathbb{C}^* = \mathbb{G}_m$, which is precisely (the image of) the maximal torus of SL_2 acting on the tangent line by the weight-2 character $t \mapsto t^2$, surjective onto $\mathbb{C}^*$.
2. **The mechanism, at every rank.** The tangent bundle of G/B is the homogeneous bundle

$$T(G/B) \cong G \times_B (\mathfrak{g}/\mathfrak{b}),$$

and the adjoint action of B on $\mathfrak{g}/\mathfrak{b}$ factors through $B \twoheadrightarrow B/U \cong T$, because the unipotent radical acts trivially on the quotient. Hence the structure group of $T(G/B)$ *canonically* reduces to the maximal torus T — for *every* G , torus or not.

3. **SL_2 is not a torus.** It is non-abelian: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ do not commute.

So “structure group reduces to a maximal torus” holds while “ G is a torus” fails: the biconditional is false. The conjecture indeed mischaracterizes triviality: $\mathcal{O}(2)$ is *not* trivial ($c_1 = 2 \neq 0$) even though its structure group is a torus. (What *is* true: $T(G/B)$ is trivial iff G is commutative, i.e. $G/B = \mathrm{pt}$ — the conjecture conflated reduction with triviality.)

3 Machine-checked discrete certificate

The identical algebraic mechanism is verified with zero axioms in the smallest faithful model $G = \mathrm{SL}_2(\mathbb{F}_2) = \mathrm{GL}_2(\mathbb{F}_2)$ ($|G| = 6 \cong S_3$; $\mathbb{F}_2^* = \{1\}$; flag set of 3 points). Matrices over $\mathbb{F}_2$ as $(a\ b; c\ d)$, $\det = ad \oplus bc$:

- $\det I = \det U = \det V = 1$ with $U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $V = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$; $UV = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = VU \Rightarrow G$ non-abelian, hence not a torus.
- $B = \{I, U\}$ (both self-inverse); $\mathfrak{g}/\mathfrak{b}$ is 1-dimensional over $\mathbb{F}_2$ with coordinate the bottom-left entry. For **all** 16 matrices $X \in \mathfrak{g}$ and both $b \in B$: $(bXb^{-1})|_{\mathfrak{g}/\mathfrak{b}} = X|_{\mathfrak{g}/\mathfrak{b}}$, so the induced transition functions of $G \times_B (\mathfrak{g}/\mathfrak{b})$ are the identity — the image of the maximal torus $T = \{\mathrm{diag}(1, 1)\} \Rightarrow$ the structure group reduces to a maximal torus.

Lean 4 (no dependencies): `Main.lean` proves `conjecture_00000001132_false : ¬ (ReducesToMaximalTorus GroupIsAbelian)` from `reduction_holds` and `not_torus`; `Check.lean` audits every theorem with `#print axioms`: all report “*does not depend on any axioms*” (no `propext`, no `Classical.choice`, no `Quot.sound`, zero `sorry`). `reproduce.py` recomputes all listed values.

4 Conclusion

Conjecture 00000001132 is DISPROVEN. The structure group of the tangent bundle of the complete flag variety reduces to a maximal torus for *every* reductive G (rank-1 example $G = \mathrm{SL}_2$, $T(G/B) \cong \mathcal{O}(2)$), so no such “iff” with “ G is a torus” can hold.